

NovaMart — Project Summary

1. Project Overview

NovaMart is a fictional retail/e-commerce business analytics project designed to simulate a real-world Data Analyst workflow. The project covered data validation, preparation, analysis, business metric development, and interactive Tableau dashboard creation.

2. Business Objectives

The objective was to understand sales performance, profitability, profit margins, customer segments, products, sellers, seller regions/types, order and payment behavior, shipping/order status, and top-performing products.

Dashboard questions: (1) How is NovaMart performing overall? (2) Which customers, products, sellers, and operational areas drive performance? (3) Where is NovaMart most profitable and which groups/products are strongest?

3. Dataset Structure

Customers

Rows: 2,000 **Columns:** 9

Key fields: Customer_ID, Customer_Name, Gender, Age, Customer_Segment, City, State, Region, Signup_Date

Products

Rows: 420 **Columns:** 7

Key fields: Product_ID, Product_Name, Category, Subcategory, Brand, Unit_Cost, Unit_Price

Sellers

Rows: 100 **Columns:** 4

Key fields: Seller_ID, Seller_Name, Seller_Type, Seller_Region

Orders

Rows: 50,000 **Columns:** 7

Key fields: Order_ID, Order_Date, Customer_ID, Seller_ID, Order_Status, Payment_Method, Shipping_Mode

Order Items

Rows: 105,953 **Columns:** 9

Key fields: Order_ID, Product_ID, Seller_ID, Quantity, Discount, Sales, Cost, Profit, Order_Status

4. Tools Used

Python, Pandas, Jupyter Notebook, SQL, and Tableau.

5. Data Analysis Workflow

- Load datasets
- Inspect structure, dimensions, columns and data types
- Check missing values and duplicates

- Profile numerical distributions
- Validate relationships between tables
- Prepare and transform data
- Calculate sales, cost, profit and profit-margin metrics
- Analyze business dimensions
- Build Tableau worksheets
- Create interactive dashboards
- Validate results against source data
- Perform final dashboard QA

6. Key Metrics

Sales

Total revenue generated from transactions.

Profit

Sales minus associated cost.

Profit Margin

Profit expressed as a percentage of sales.

7. Tableau Dashboards

Dashboard 1 — Executive Overview

- Total Profit
- Total Sales
- Profit Margin
- Monthly Sales & Profit Trend
- Sales by Region
- Sales by Product Category

Dashboard 2 — Detailed Performance

- Sales by Customer Segment
- Profit Margin by Customer Segment
- Sales by Product Subcategory
- Profit Margin by Product Subcategory
- Sales by Seller
- Sales by Seller Region
- Profit by Seller Region
- Order & Payment Analysis

- Shipping & Order Status Analysis

Dashboard 3 — Profitability & Top Performers

- Profit Margin Trend
- Profit by Product Category
- Profit by Product Subcategory
- Top Products by Sales
- Top Products by Profit Margin
- Sales by Seller Type
- Profit Margin by Seller Region

8. Dashboard Design

The dashboards were separated by business purpose rather than placing every visualization on one page. KPI cards were used for headline metrics, while detailed comparisons remained charts. Consistent typography, blue visual elements, compact number formatting, structured containers, appropriate filters, and controlled scrolling were used to improve readability.

9. Key Observations

- Approximately ■2.107 billion in sales
- Approximately ■635.1 million in profit
- Overall profit margin approximately 30.1%
- Regional performance differs across sales and profitability
- Seller performance varies significantly
- Product categories/subcategories differ in profit and margin
- Highest-sales products are not necessarily highest-margin products
- Customer segments differ in sales and profitability
- Payment, shipping and order-status patterns provide additional operational insight

10. Challenges & Solutions

Multiple related datasets

Validated table structures and relationships before cross-table analysis.

Dashboard duplication

Separated dashboards by purpose so KPIs, comparisons and detailed analysis were not unnecessarily duplicated.

High-cardinality dimensions

Retained scrolling where appropriate instead of compressing charts into unreadable visuals.

Layout and readability

Used containers, consistent spacing, font hierarchy, compact number formatting and cleaner KPI cards.

Consistency

Performed a final QA pass across all dashboards.

11. Final Outcome

The project progressed from raw multi-table retail data to a complete business intelligence solution containing validated datasets, Python/Pandas analysis, SQL analysis, business metrics, Tableau worksheets, three purpose-specific dashboards, and final QA.

12. Portfolio Summary

NovaMart is an end-to-end retail analytics project demonstrating the ability to work with relational datasets, validate and prepare data, analyze business performance using Python and SQL, and communicate findings through interactive Tableau dashboards.